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**PRN - B24CE1079**

**Subject - Object Oriented Programming**

**Assignment 8**

/\*PROBLEM STATEMENT: Design a program with a template for sorting the accepted array and displaying it using integer or float type data. Implement any sorting type using Generic Programming.\*/

**PROGRAM**

```
#include <iostream>
using namespace std;

// Template function to sort the array
template <class T>
void sortArray(T arr[], int n) {
    for (int i = 0; i < n - 1; i++) {
        int pos = i;
        for (int j = i + 1; j < n; j++) {
            if (arr[j] < arr[pos])
                pos = j;
        }
        T temp = arr[i];
        arr[i] = arr[pos];
        arr[pos] = temp;
    }
}

// Template function to display the array
template <class T>
void displayArray(T arr[], int n) {
    for (int i = 0; i < n; i++)
        cout << arr[i] << " ";
    cout << endl;
}

int main() {
    int choice;
    int n;

    cout << "Enter number of elements: ";
    cin >> n;
```

```

do {
    cout << "\nChoose data type:\n";
    cout << "1. Integer\n";
    cout << "2. Float\n";
    cout << "3. Exit\n";
    cout << "Enter your choice: ";
    cin >> choice;

    switch (choice) {
        case 1: {
            int arr[50];
            cout << "Enter " << n << " integer elements: ";
            for (int i = 0; i < n; i++)
                cin >> arr[i];
            sortArray(arr, n);
            cout << "Sorted array (Integers): ";
            displayArray(arr, n);
            break;
        }

        case 2: {
            float arr[50];
            cout << "Enter " << n << " float elements: ";
            for (int i = 0; i < n; i++)
                cin >> arr[i];
            sortArray(arr, n);
            cout << "Sorted array (Floats): ";
            displayArray(arr, n);
            break;
        }

        case 3:
            cout << "\nExiting program..." << endl;
            break;

        default:
            cout << "\nInvalid choice! Please try again." << endl;
    }

} while (choice != 3);

return 0;
}

```

## OUTPUT

```
Choose data type:
1. Integer
2. Float
3. Exit
Enter your choice: 1
Enter 5 integer elements: 9 3 6 2 1
Sorted array (Integers): 1 2 3 6 9

Choose data type:
1. Integer
2. Float
3. Exit
Enter your choice: 2
Enter 5 float elements: 12.5 9.8 45.6 3.4 2.7
Sorted array (Floats): 2.7 3.4 9.8 12.5 45.6

Choose data type:
1. Integer
2. Float
3. Exit
Enter your choice: 3

Exiting program...
```